SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	G.Hima Bindu
Register Number	192511377

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)
Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Duration : 60 Minutes

Code of Conduct
I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Hima Bindu

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM:

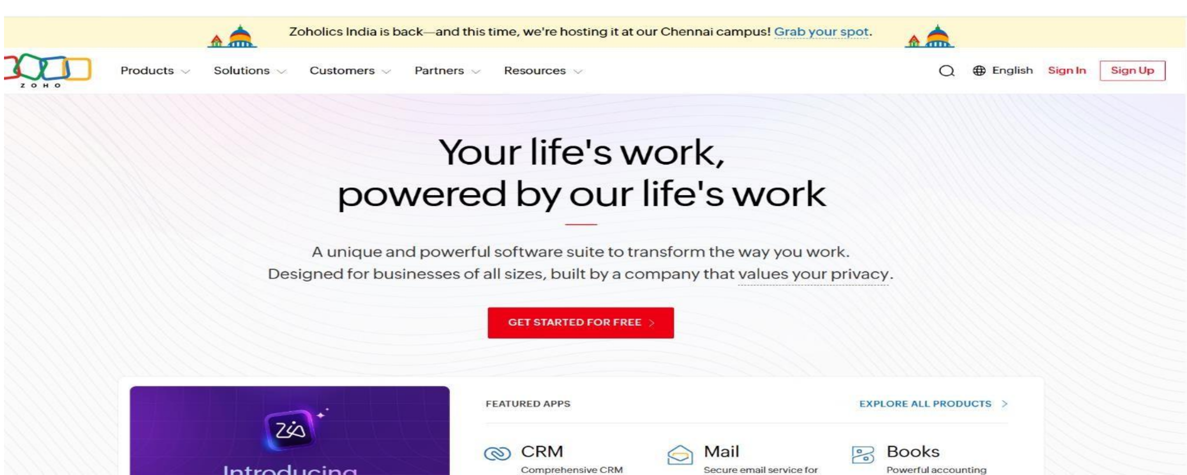
To develop a simple cloud-based Library Book Reservation System for SIMATS Library using a Cloud Service Provider and demonstrate the Software as a Service (SaaS) model by managing book details, availability, and reservations.

ALGORITHM:

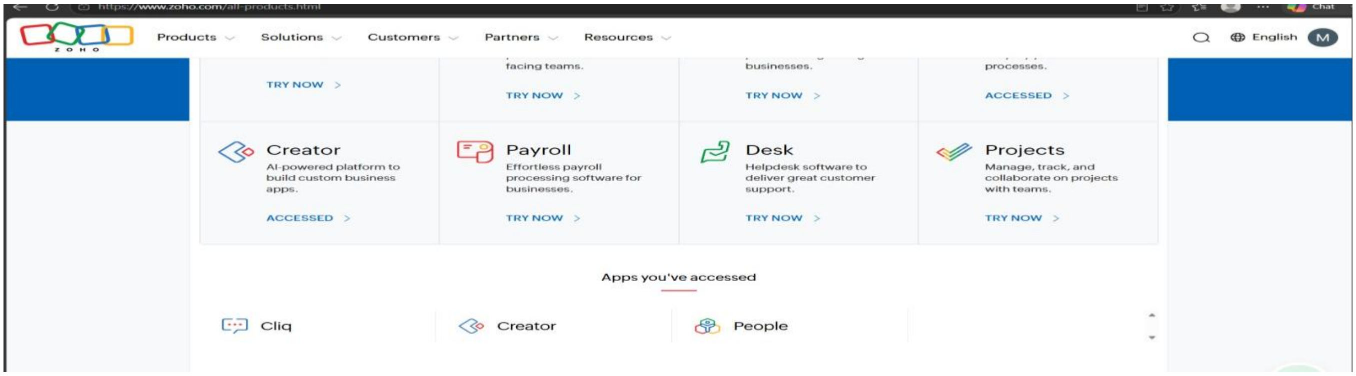
1. Start
2. Open the Library Book Reservation System.
3. Enter the book details such as Title, Author, Publication, Year, Edition, No. of Copies, Rack No., and Status.
4. Store the details in the cloud database.
5. Search for the required book.
6. Check whether the book is Available/Taken.
7. If available, reserve the book and update the status as Taken.
8. If the book is returned, update the status as Returned/Available.
9. Display the updated book details.
10. Stop.

Implemented:

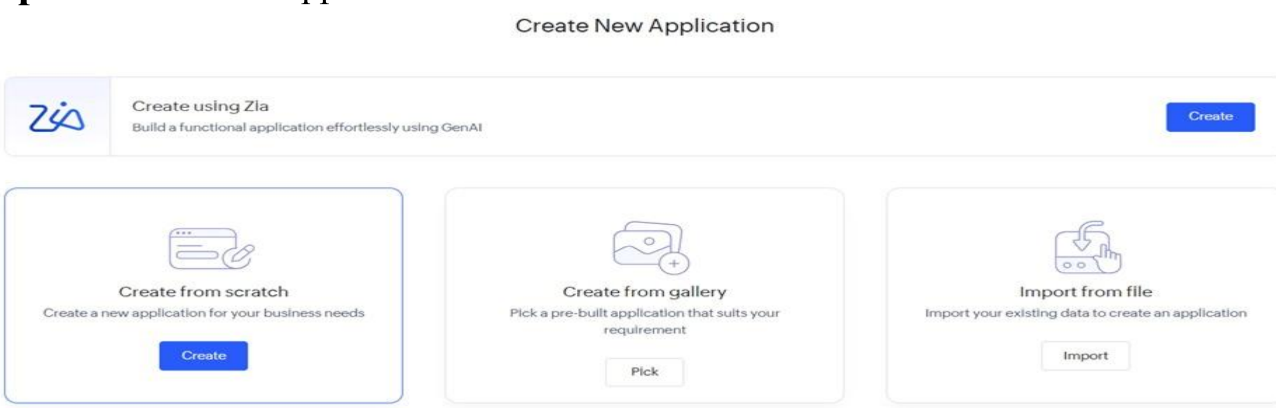
Step1: Go to Zoho .com



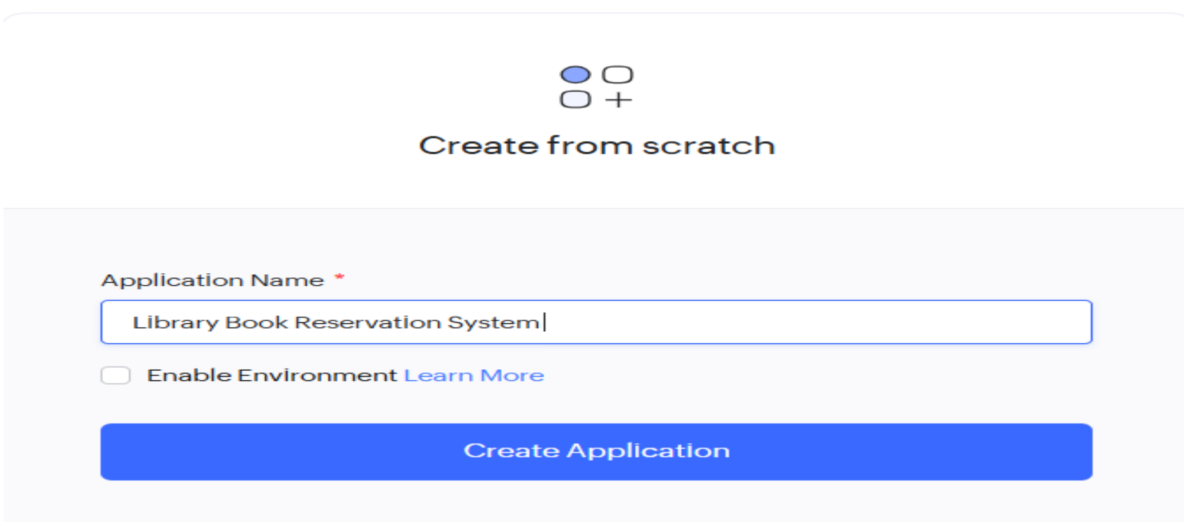
Step2: Login to Zoho.Com



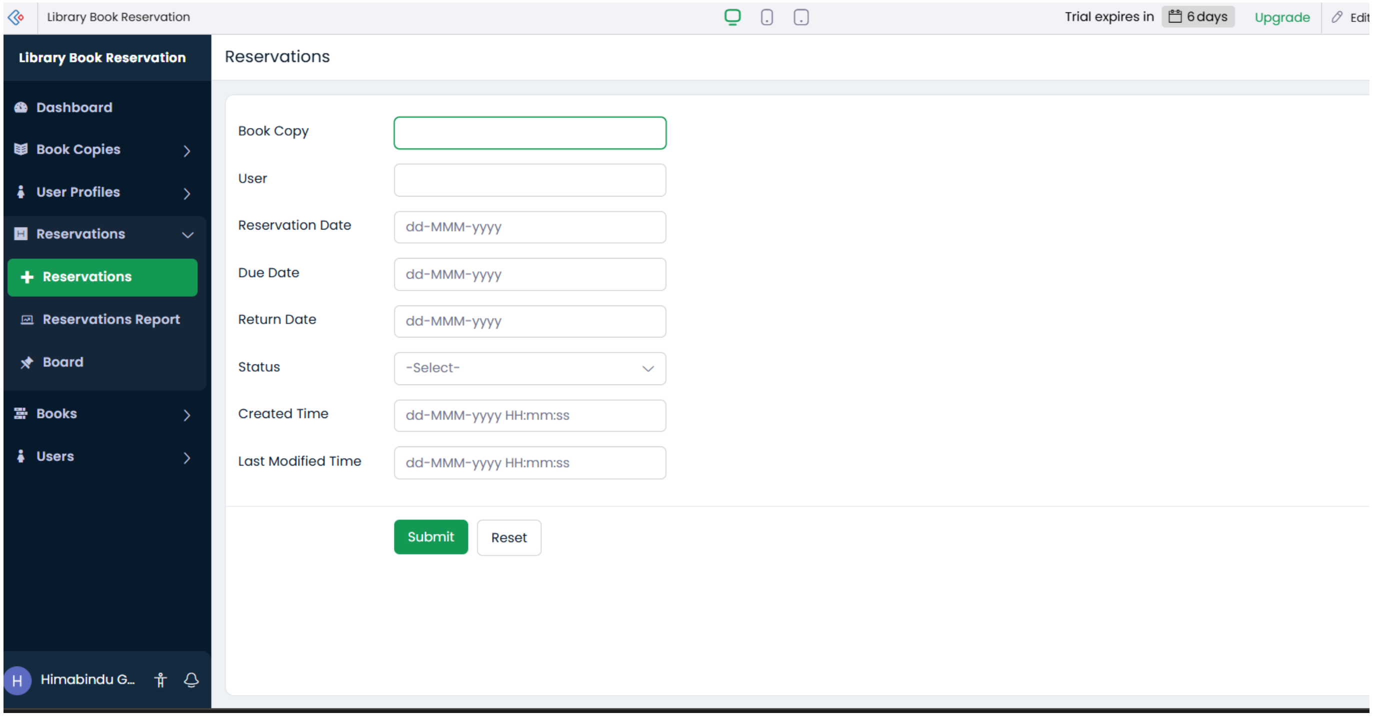
Step3: Create New Application from Select from Scratch



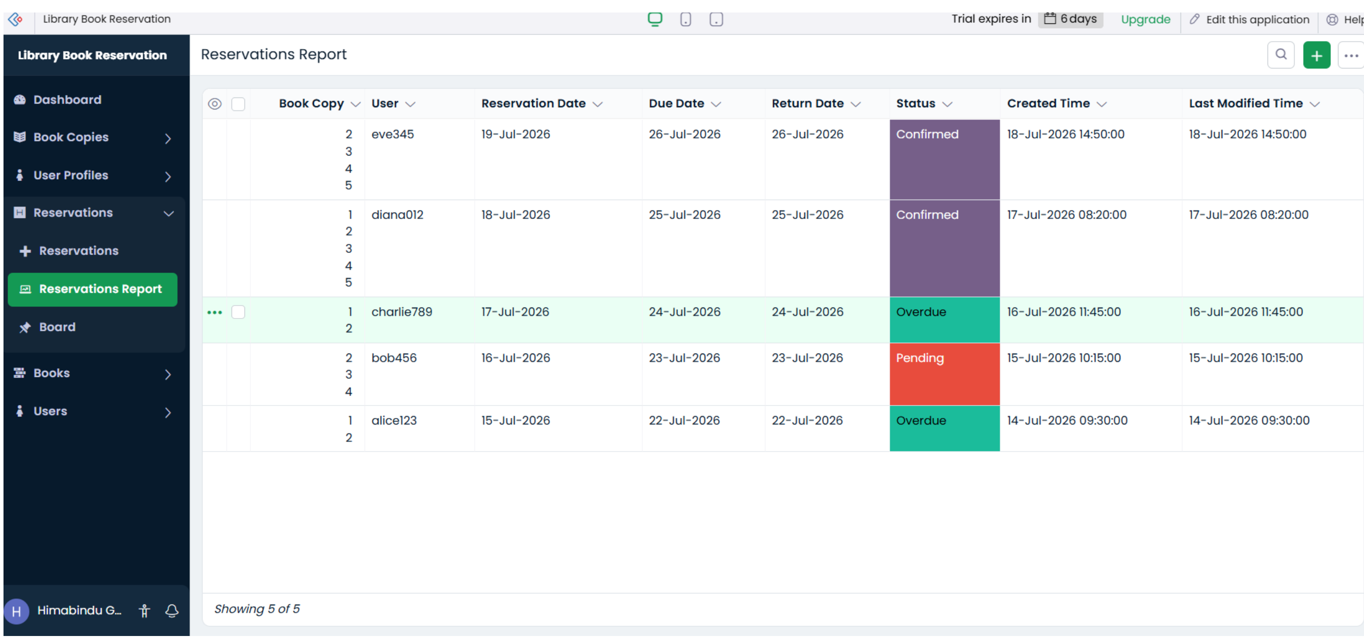
STEP 4: Enter Library Book Reservation System as the application name and click Create Application**.**

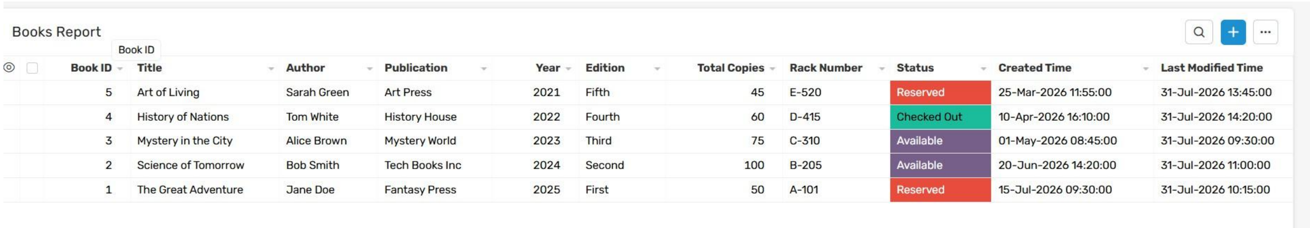


Step 5: Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status



STEP 6: Save the form, enter sample booking records, and view them in the Library Book Reservation System.

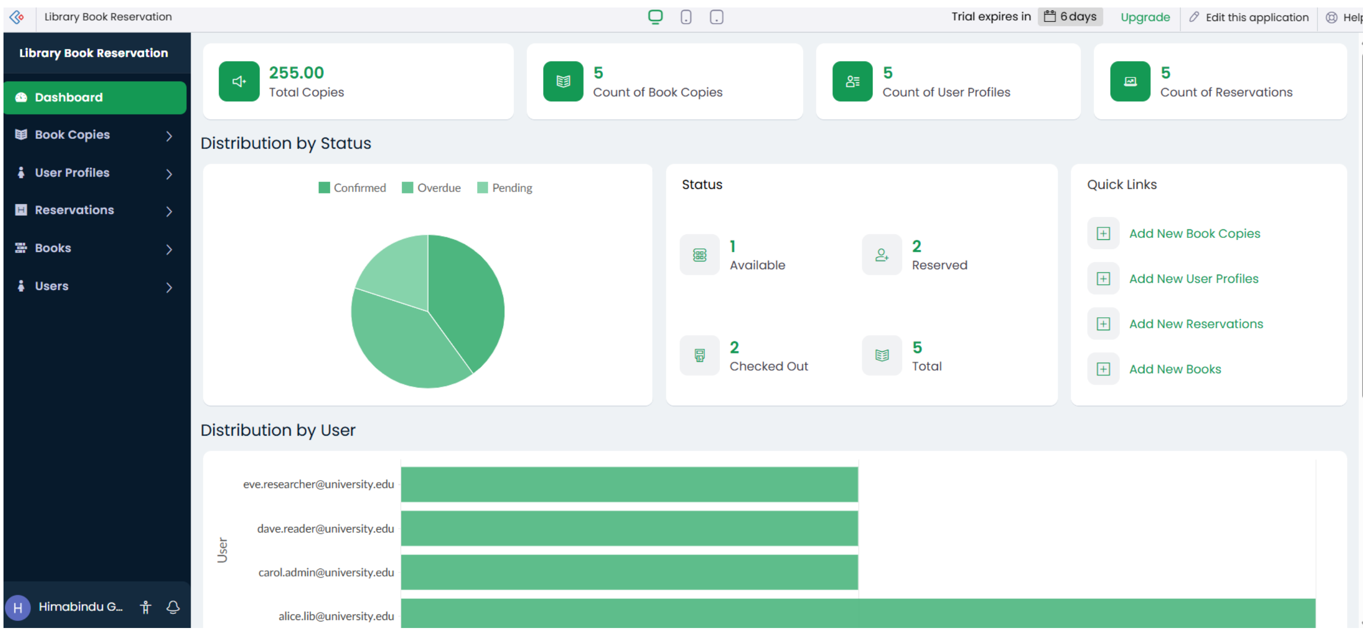




Books Report

Book ID	Title	Author	Publication	Year	Edition	Total Copies	Rack Number	Status	Created Time	Last Modified Time
5	Art of Living	Sarah Green	Art Press	2021	Fifth	45	E-520	Reserved	25-Mar-2026 11:55:00	31-Jul-2026 13:45:00
4	History of Nations	Tom White	History House	2022	Fourth	60	D-415	Checked Out	10-Apr-2026 16:10:00	31-Jul-2026 14:20:00
3	Mystery in the City	Alice Brown	Mystery World	2023	Third	75	C-310	Available	01-May-2026 08:45:00	31-Jul-2026 09:30:00
2	Science of Tomorrow	Bob Smith	Tech Books Inc	2024	Second	100	B-205	Available	20-Jun-2026 14:20:00	31-Jul-2026 11:00:00
1	The Great Adventure	Jane Doe	Fantasy Press	2025	First	50	A-101	Reserved	15-Jul-2026 09:30:00	31-Jul-2026 10:15:00

RESULT: The software is been created



Thus, the SIMATS Library Book Reservation System was successfully designed as a cloud-based SaaS application. The system can store and manage book details, check availability, reserve books, and update the status of books such as Taken, Returned, and Available through a web-based interface.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM:

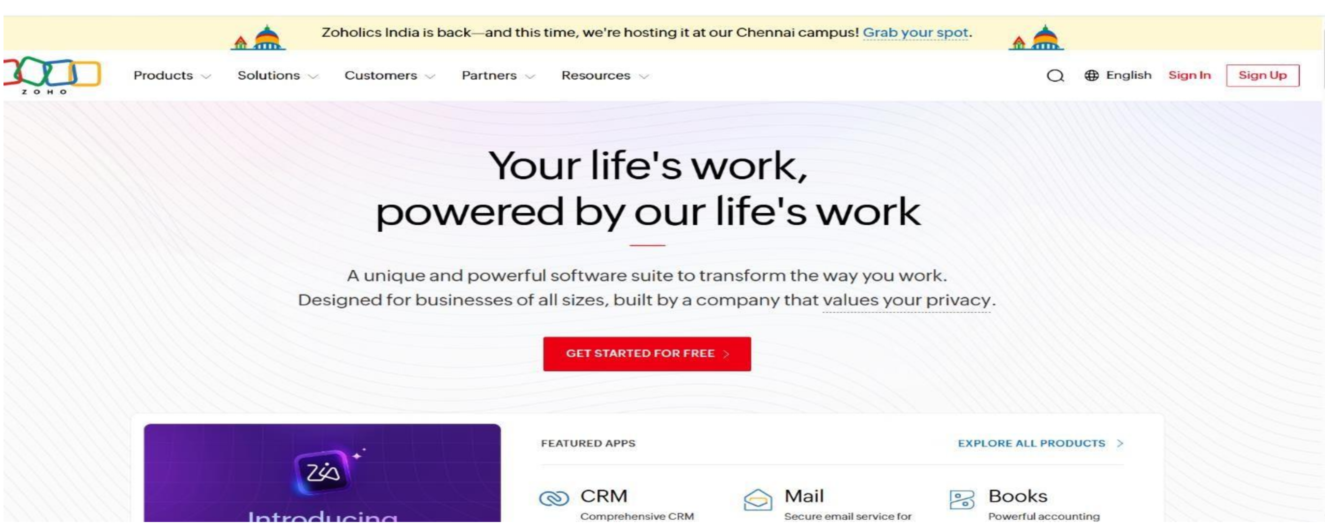
To develop a simple cloud-based Payroll Processing System using a Cloud Service Provider and demonstrate the Software as a Service (SaaS) model by managing employee details, salary components, and calculating gross and net salary.

ALGORITHM:

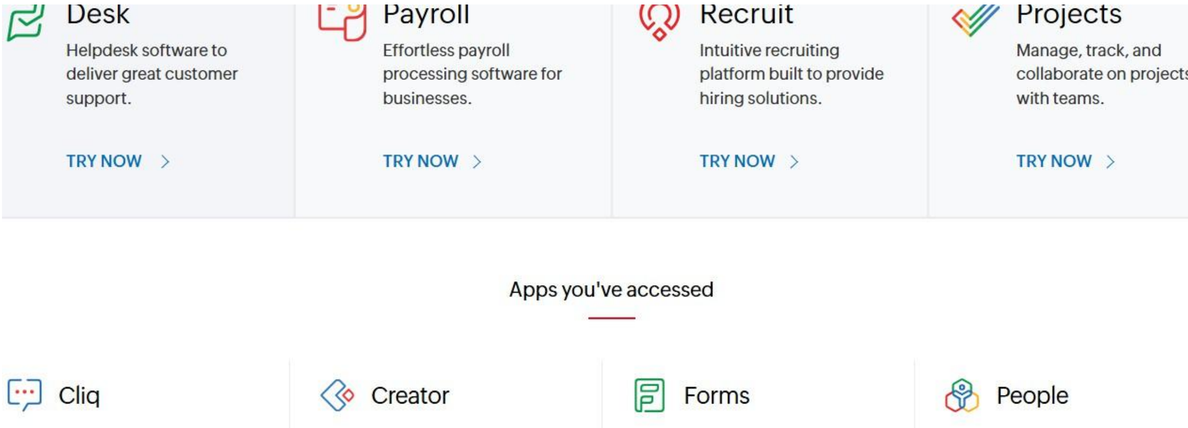
1. Start
2. Open the Payroll Processing System.
3. Enter employee details such as Name, Employee ID, Present Organization Experience, and Overall Experience.
4. Enter salary details such as Basic Pay, HRA, DA, and TA.
5. Calculate Gross Pay by adding Basic Pay, HRA, DA, and TA.
6. Calculate deductions, if applicable.
7. Calculate Net Pay after subtracting deductions from Gross Pay.
8. Store the employee payroll details in the cloud database.
9. Display the employee salary details and Net Pay.
10. Stop.

Implemented:

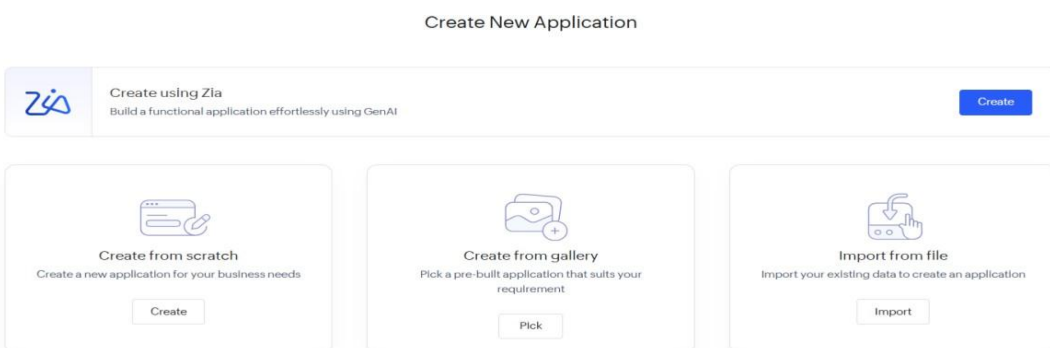
Step1: Go to Zoho.com



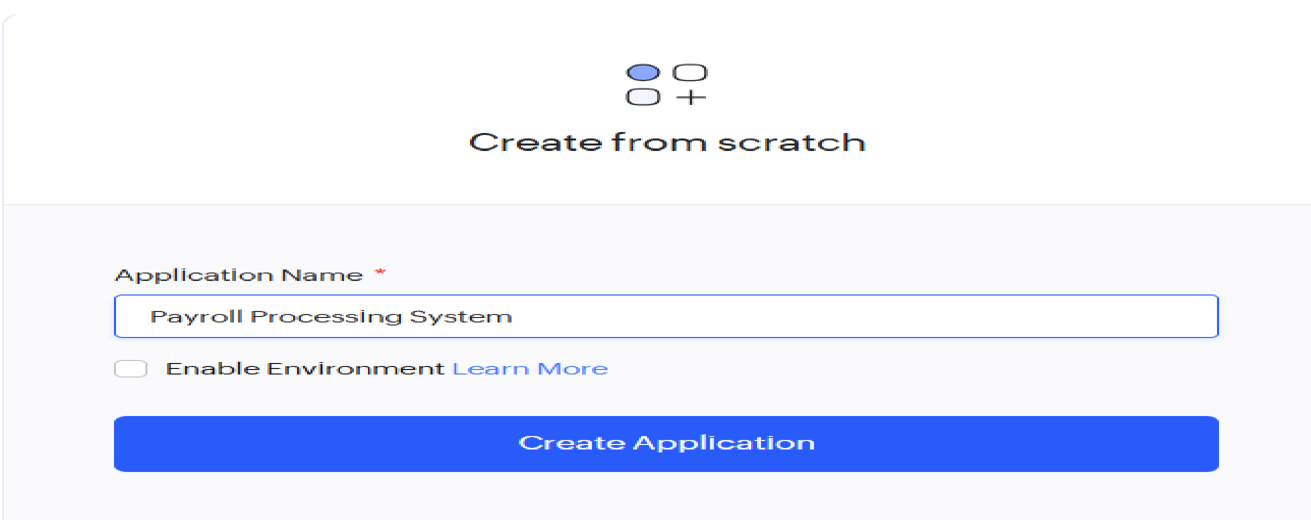
Step2: Login to your zoho account and open your zoho account



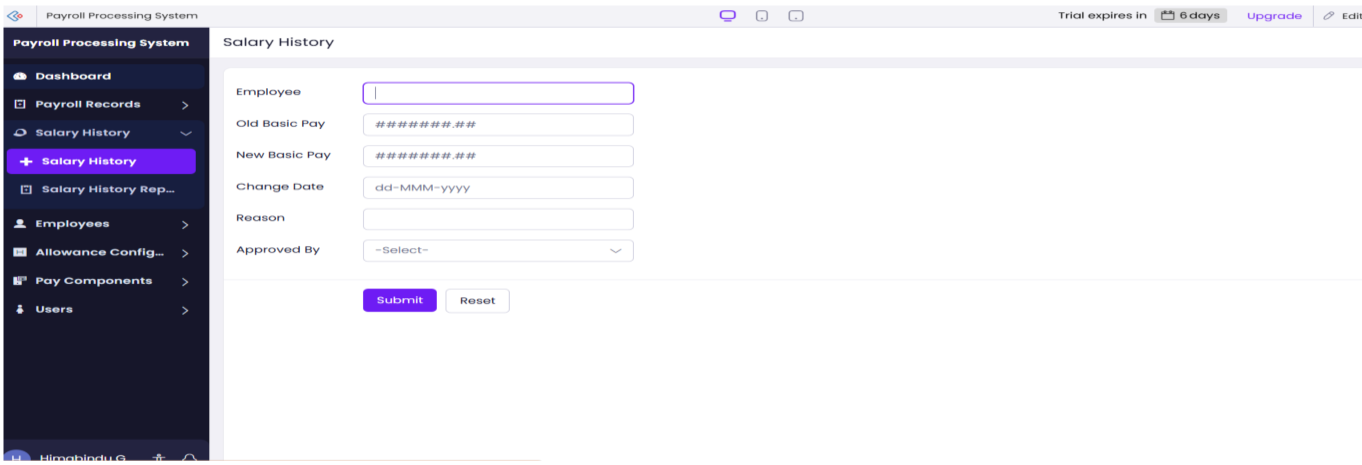
Step3: Click Create New Application and select Create from Scratch

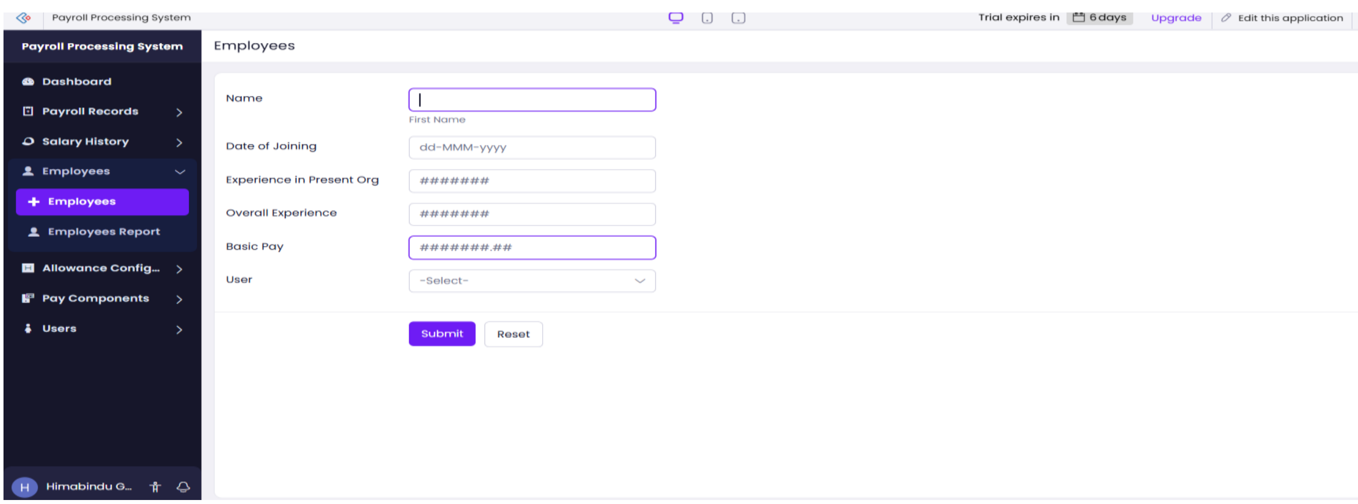


Step4: Enter payroll processing system as a application name and click on create application

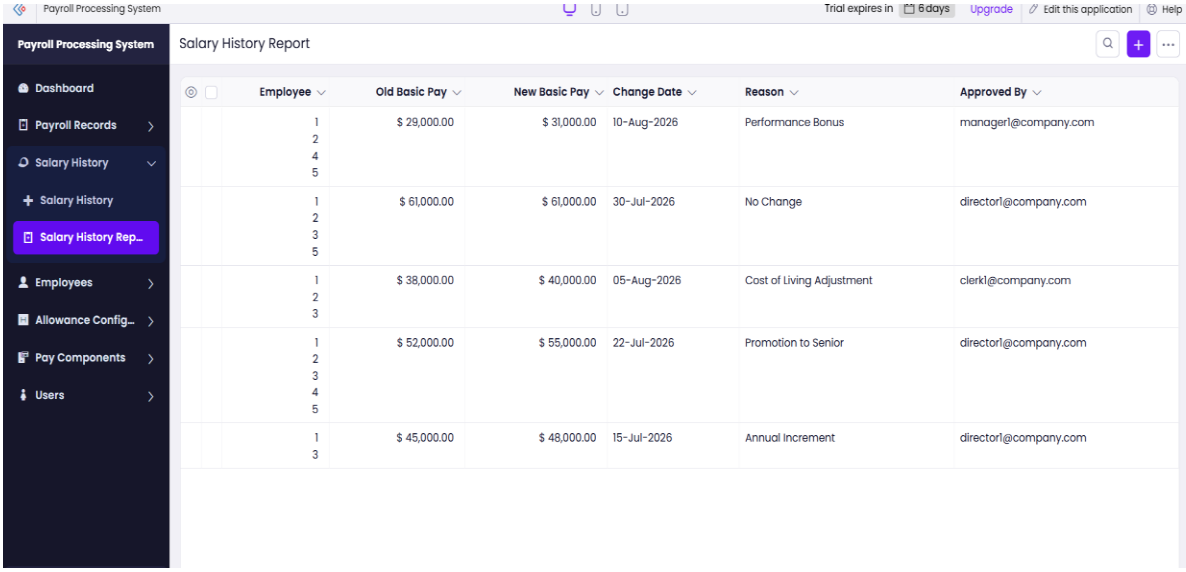


Step5: : Add fields: Student Name, Register Number, Subject Marks, Total, Average, Percentage, and Rank.

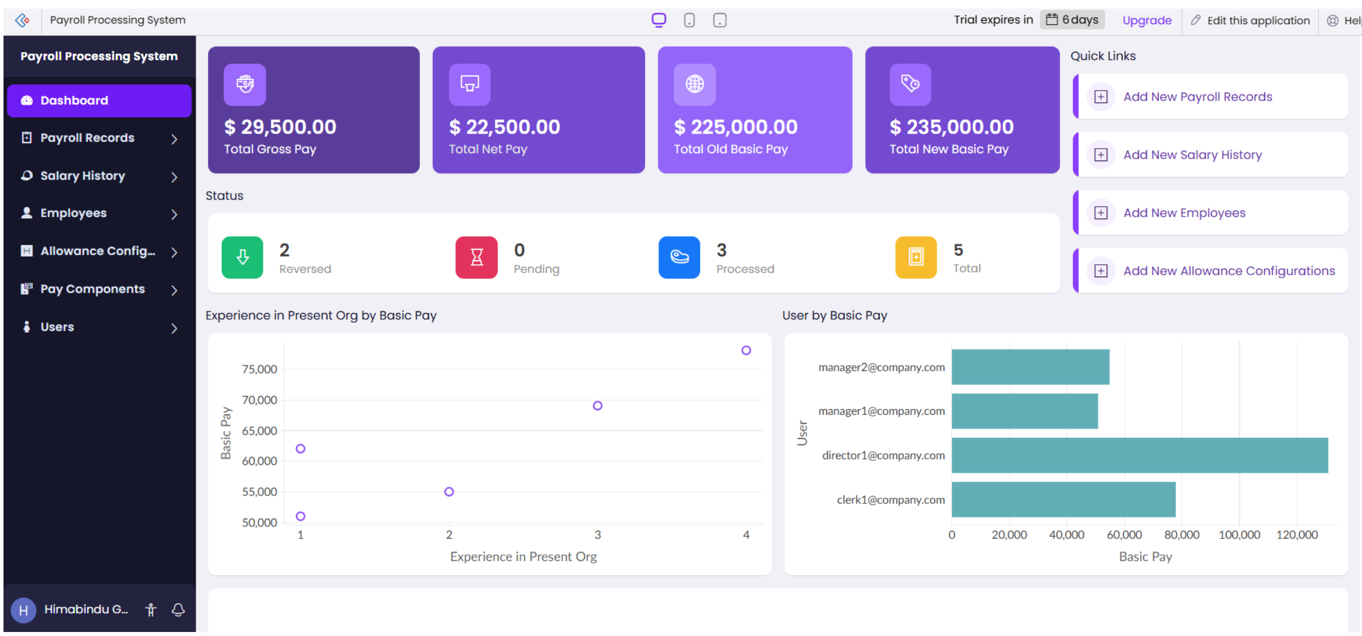




Step6: Save the form payroll processing ,enter sample booking records and view in the system



RESULT: The Software has been created



Thus, the cloud-based Payroll Processing System was successfully designed as a SaaS application. The system stores employee information in the cloud and calculates HRA, DA, TA, Gross Pay, and Net Pay efficiently. The payroll details can be accessed and managed through a web-based application from different device.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

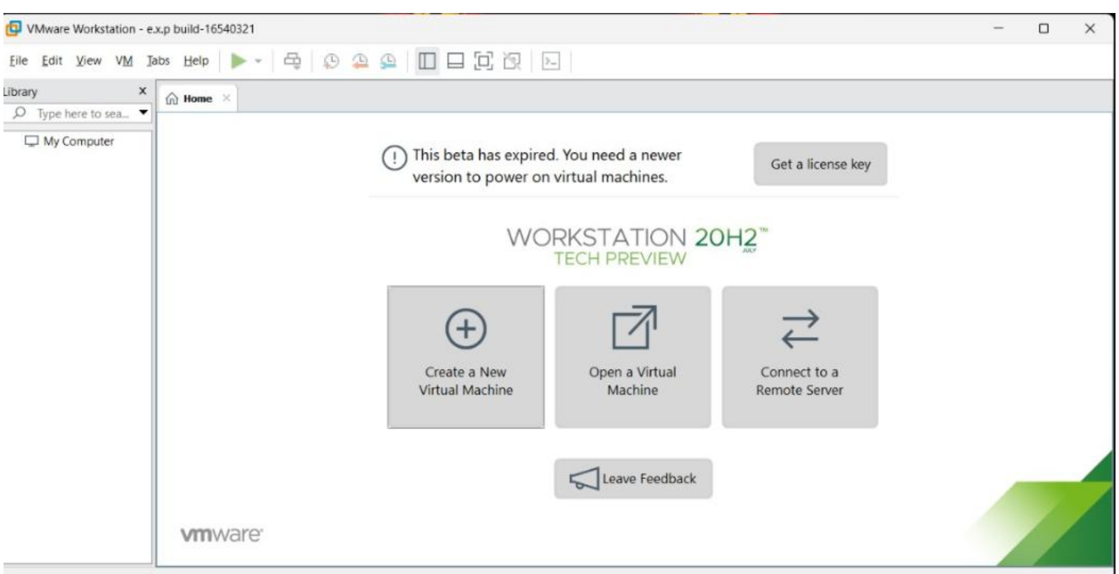
To demonstrate virtualization by installing a Type-2 Hypervisor (VMware Workstation) and creating a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage.

Steps:

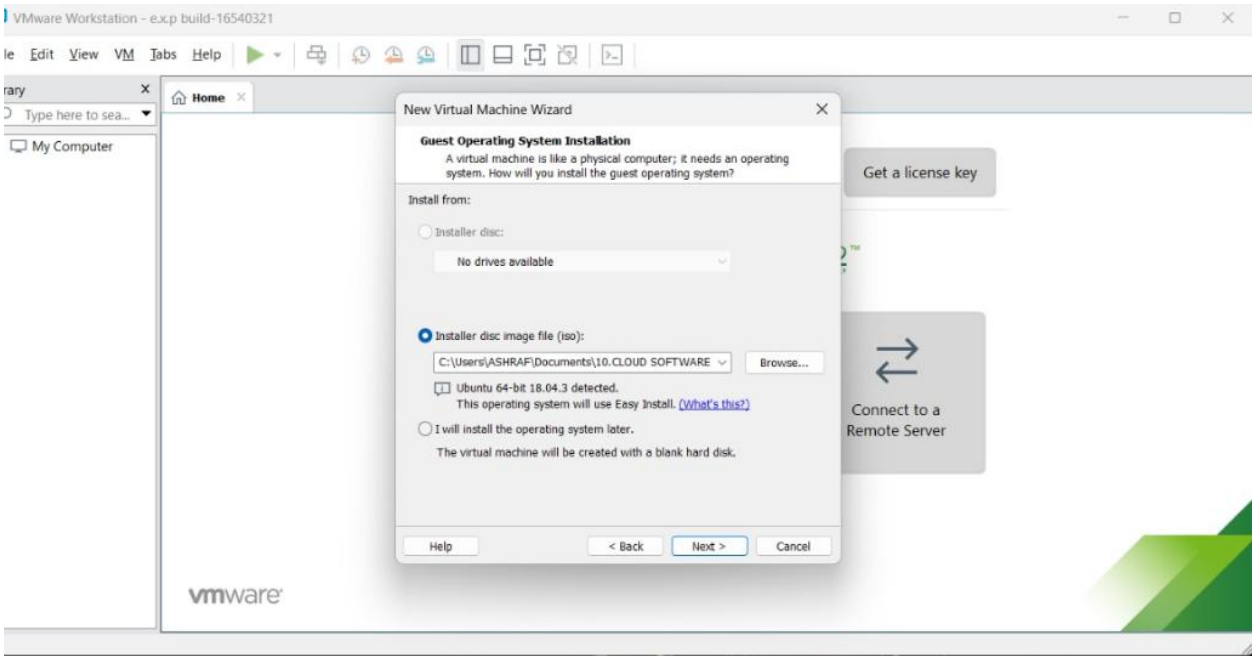
1. Open VMware and select Create New Virtual Machine.
2. Select the Ubuntu ISO file.
3. Set the VM name as Ubuntu-VM.
4. Set RAM = 2 GB and CPU = 1.
5. Set the hard disk size = 15 GB.
6. Start the VM and install Ubuntu.

Implemented:

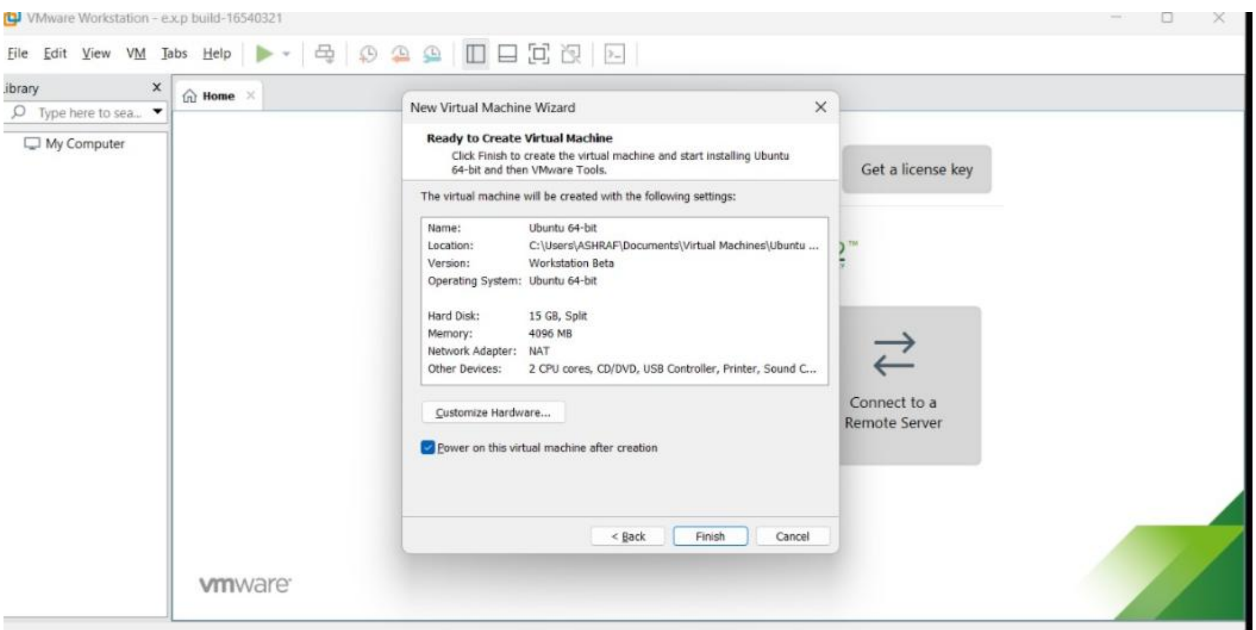
1. Open VMware and select Create New Virtual Machine.



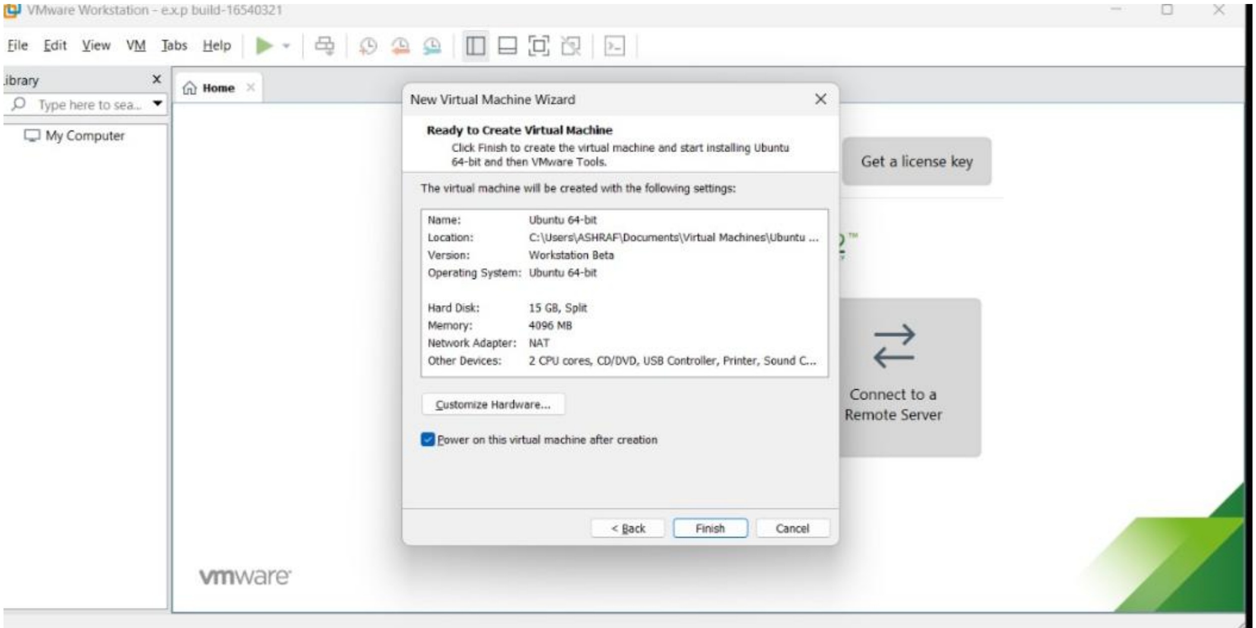
2. Select the Ubuntu ISO file.



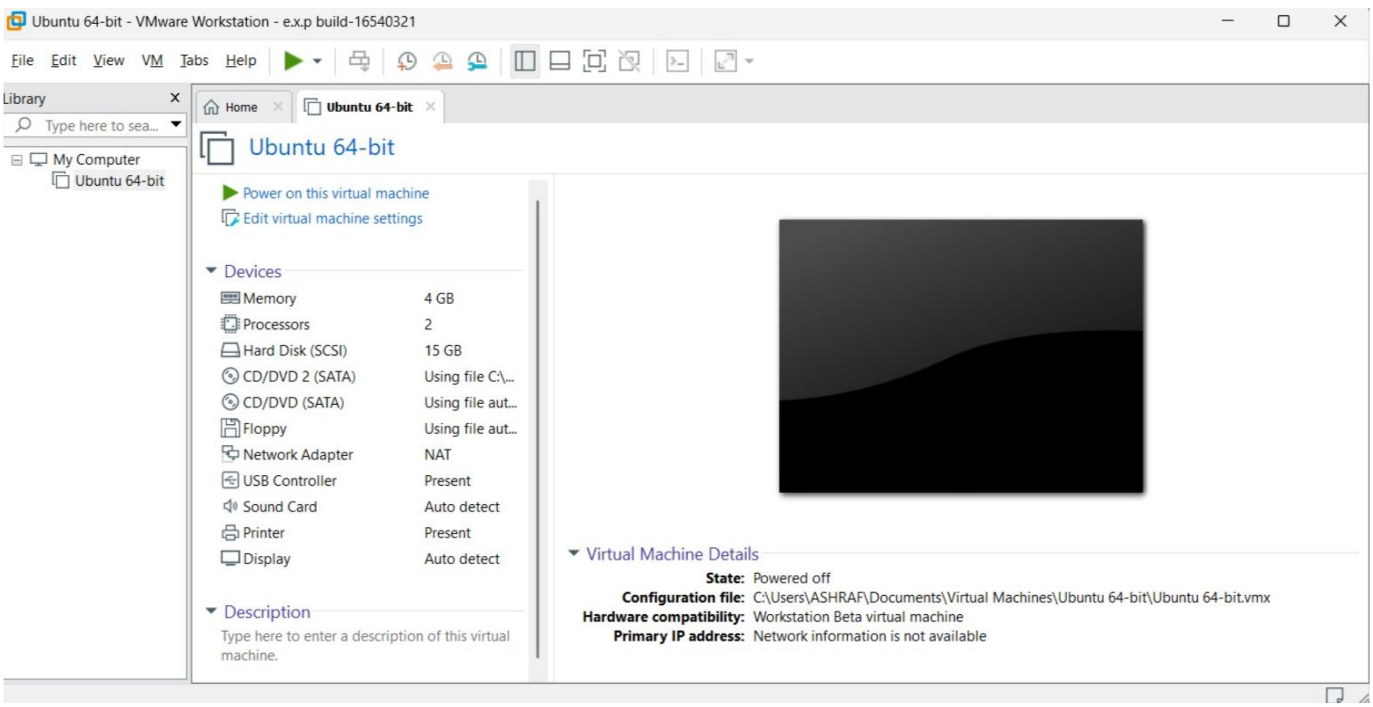
3.Set the VM name as Ubuntu-VM.



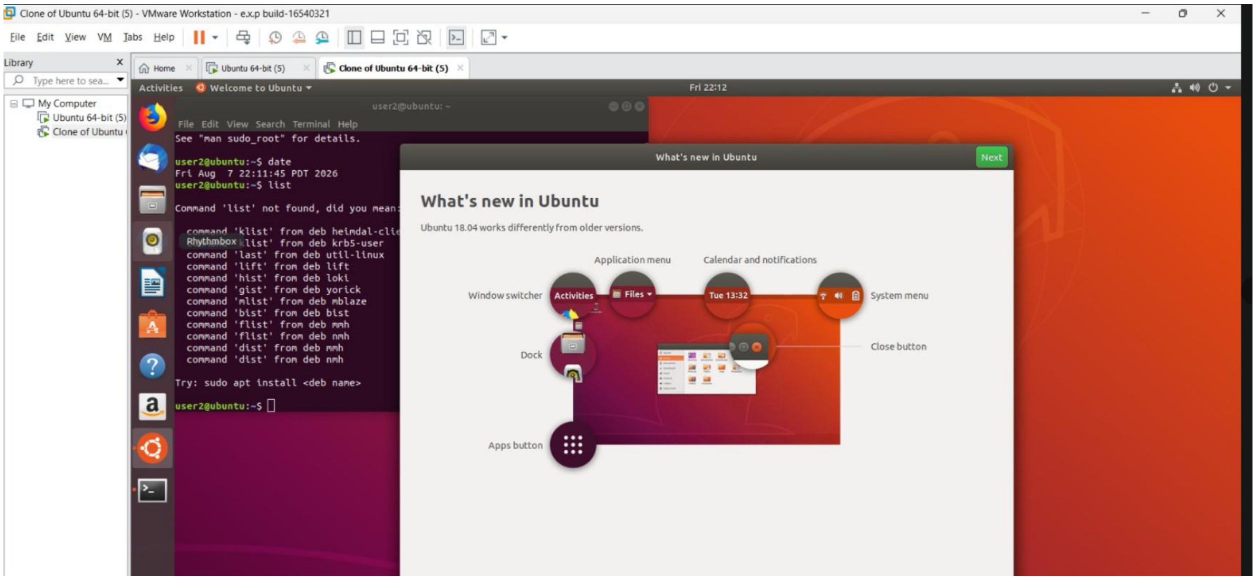
4.Set RAM = 2 GB and CPU = 1.



.5 Set the hard disk size = 15 GB.



6.Start the VM and install Ubuntu.



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim

To demonstrate **virtualization** by installing a **Type-2 Hypervisor (VMware Workstation)**, creating a Virtual Machine with a Windows/Linux operating system, and **cloning the VM** to create and test a previous version of the Virtual Machine.

Steps

.1 Open VMware and select Create New Virtual Machine.

2. Select the Ubuntu ISO file.

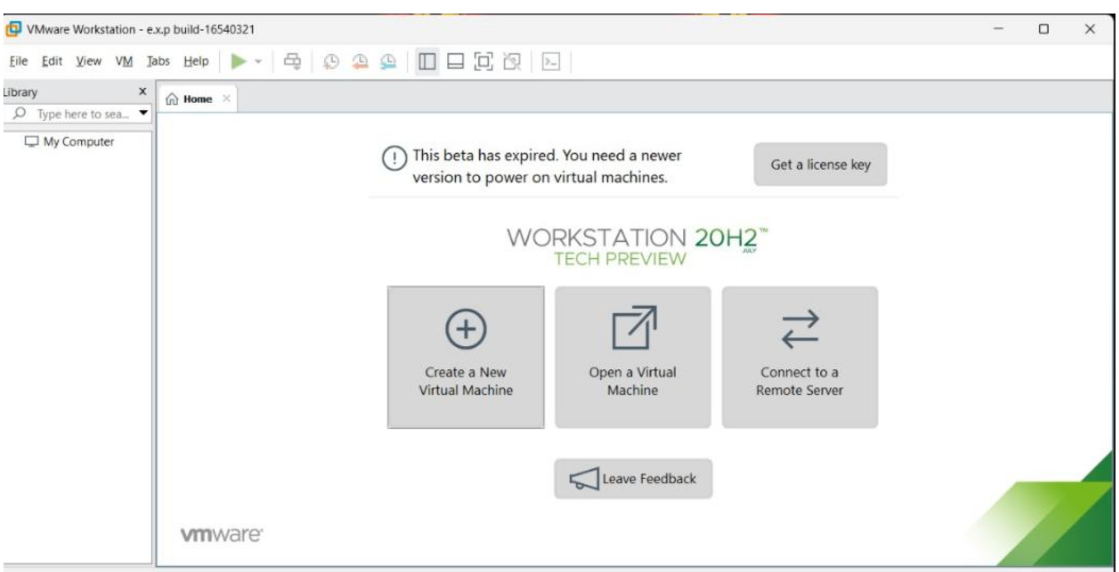
3. Set the VM name as Ubuntu-VM.

4. Set RAM = 2 GB and CPU = 1.

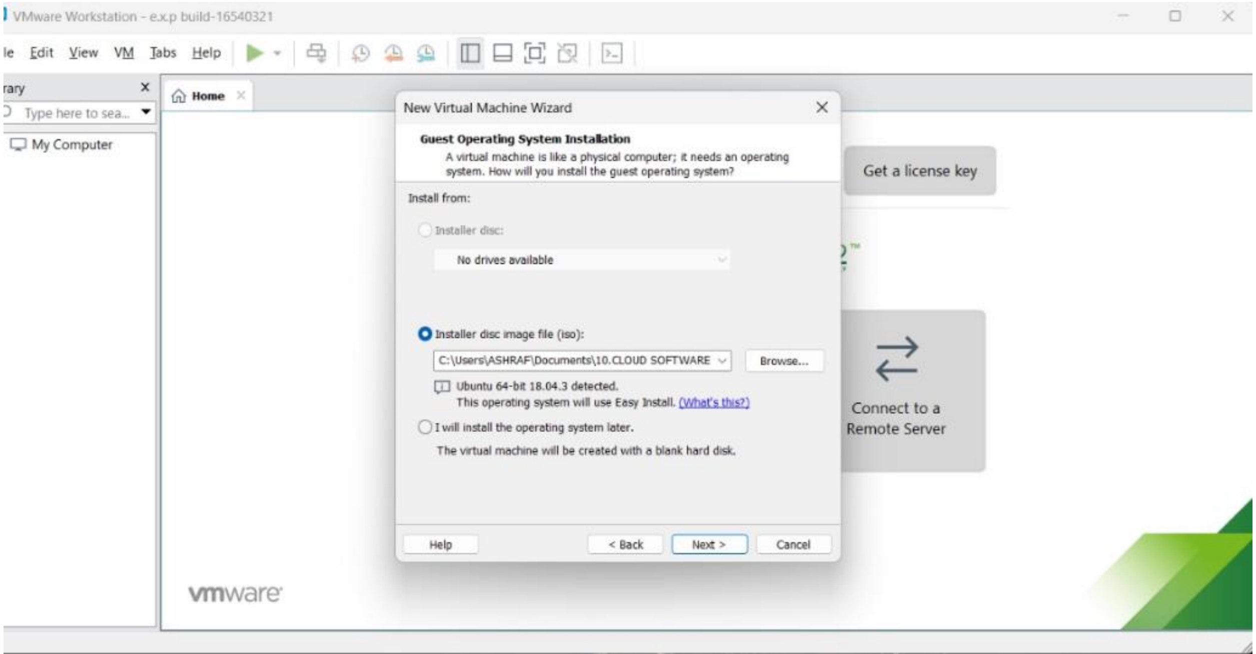
5. Set the hard disk size = 15 GB.

6. Start the VM and install Ubuntu.

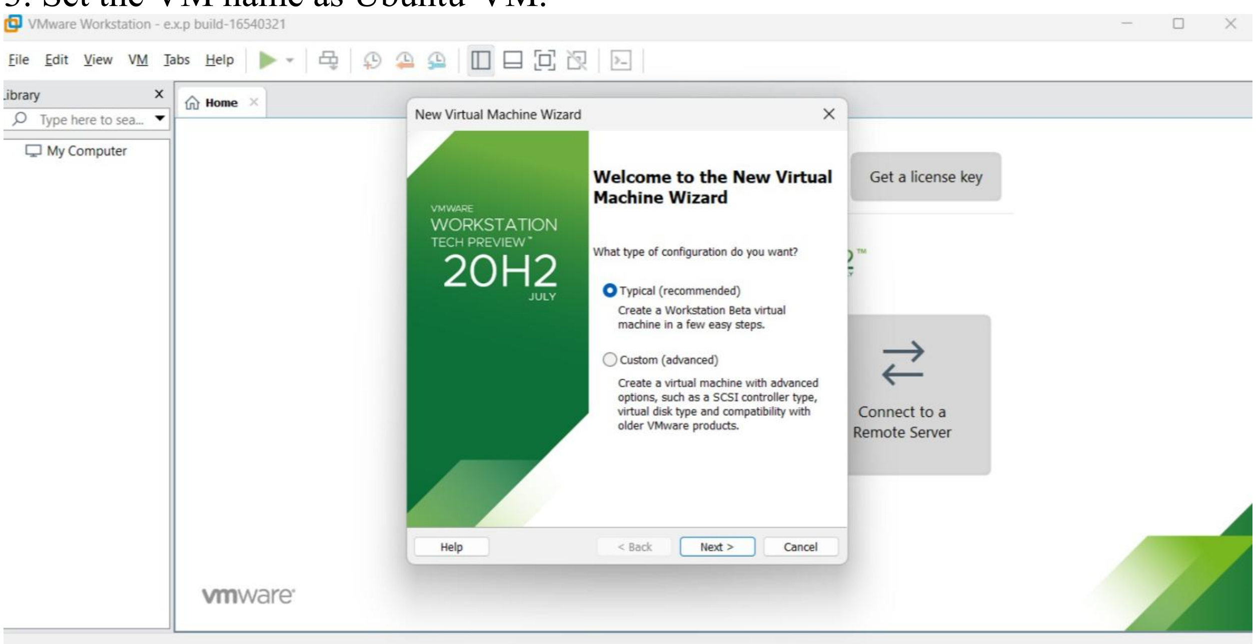
1 Open VMware and select Create New Virtual Machine.



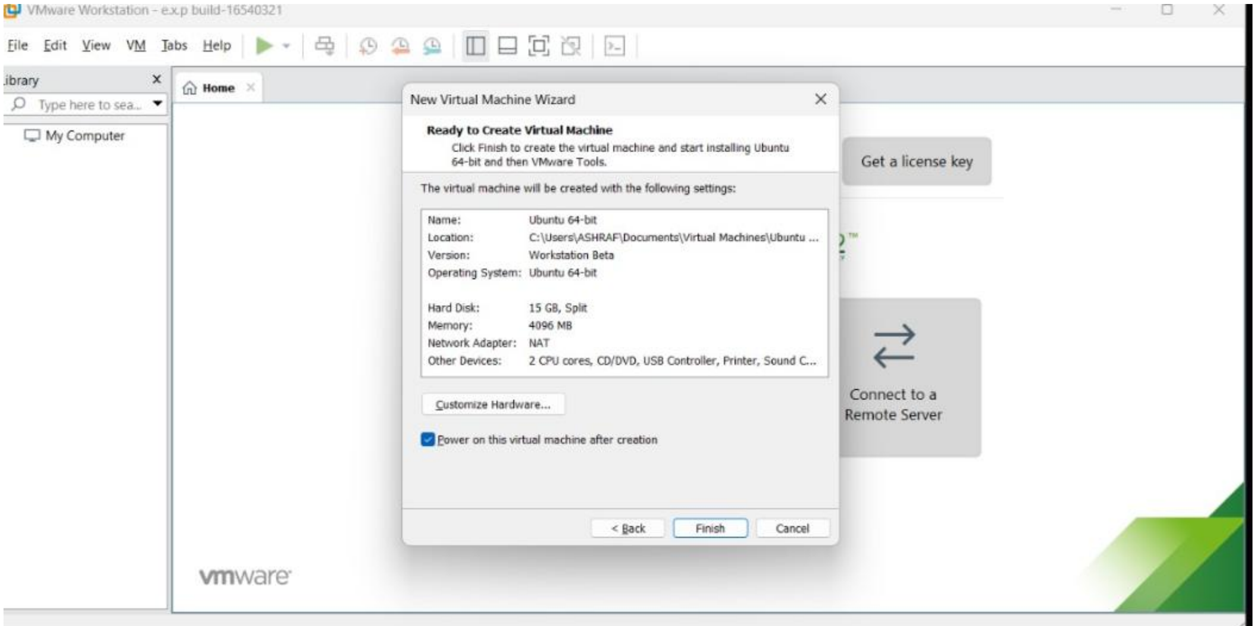
2. Select the Ubuntu ISO file.



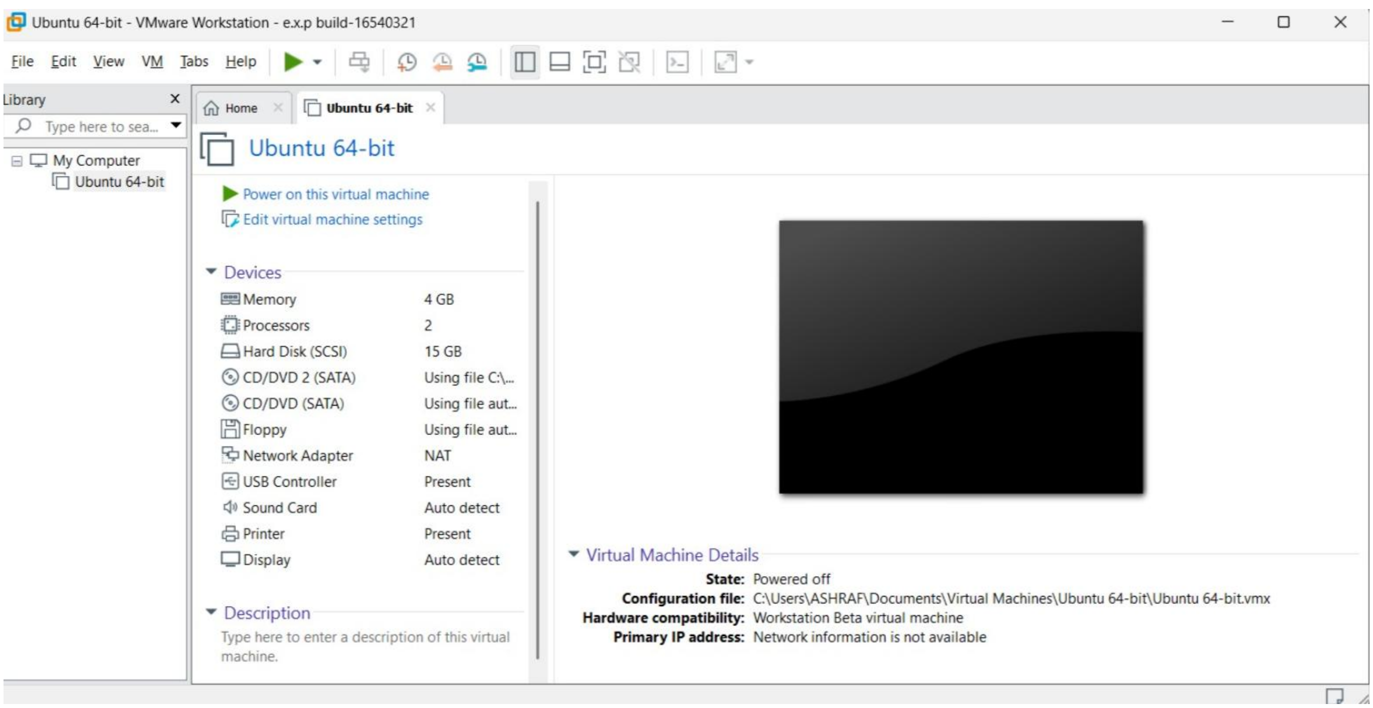
3. Set the VM name as Ubuntu-VM.



4.Set RAM = 2 GB and CPU = 1.



.5 Set the hard disk size = 15 GB.



6.Start the VM and install Ubuntu.